

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Sequences & Series

Mixed series questions are about recognising the structure before choosing the formula.

SEQUENCES & SERIES · P2 · 07

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

01 SPOT: Spot The Type

TRIGGER *sequence or series unknown type*

02 SIGMA: Sigma Notation

TRIGGER Σ *notation*

03 LINK: Link Term And Sum

TRIGGER *given S_n , find u_n*

04 MIX: Mixed Series Equation

TRIGGER *series condition, unknown n*

BANK EVIDENCE Local pages: 17 problem, 35 answer, 12 notes. Website primary entries: 10.

01 SPOT: Spot The Type

TRIGGER *sequence or series unknown type*

BECOMES Check difference and ratio.

FIRST LINE TO WRITE

$$d = u_{n+1} - u_n, \quad r = \frac{u_{n+1}}{u_n}$$

SIMPLEST STRATEGY

- 1 Scan consecutive terms.
- 2 Probe difference.
- 3 Observe ratio.
- 4 Take the matching formula.

WORKED MODEL

| 2, 6, 18 has ratio 3, so it is geometric.

HARD VARIANTS

- 1 Some hard questions hide terms as algebraic expressions; simplify first.
- 2 If neither constant difference nor ratio appears, do not force a formula.
- 3 State the type before applying a sum formula.

— BOTTOM LINE

The type decides the formula.

SAME IDEA Use SPOT: Spot The Type.

QUESTION

Decide whether $5, 9, 13, \dots$ is arithmetic or geometric.

YOUR SOLUTION

02 SIGMA: Sigma Notation

TRIGGER Σ notation

BECOMES Read lower limit, upper limit, and term formula.

FIRST LINE TO WRITE

$$\sum_{r=1}^n u_r$$

SIMPLEST STRATEGY

- 1 Start at the lower limit.
- 2 Insert the term expression.
- 3 Generate the first few terms if needed.
- 4 Match to known sum.
- 5 Add using formula.

WORKED MODEL

$$\sum_{r=1}^n (2r + 1) = 2 \sum r + \sum 1.$$

HARD VARIANTS

- 1 Split sigma sums term by term.
- 2 Use standard $\sum r$ and $\sum r^2$ only when available in the course.
- 3 If the lower limit is not 1, subtract the earlier terms.

— BOTTOM LINE

Sigma is compressed summation, not a new topic.

03 LINK: Link Term And Sum

TRIGGER *given S_n , find u_n*

BECOMES A term is the difference of two consecutive sums.

FIRST LINE TO WRITE

$$u_n = S_n - S_{n-1}$$

SIMPLEST STRATEGY

- 1 Locate the sum formula.
- 2 Insert n and $n - 1$.
- 3 Net the difference.
- 4 Keep the final term expression.

WORKED MODEL

| If $S_n = n^2$, then $u_n = n^2 - (n - 1)^2 = 2n - 1$.

HARD VARIANTS

- 1 Check $u_1 = S_1$ after deriving the formula.
- 2 If S_n is quadratic, u_n is usually linear.
- 3 Use the term formula to test arithmetic or geometric behaviour.

— BOTTOM LINE

Terms come from changes in sums.

04 MIX: Mixed Series Equation

TRIGGER *series condition, unknown n*

BECOMES Build the equation from the correct sum formula.

FIRST LINE TO WRITE

$$S_n = \text{given value}$$

SIMPLEST STRATEGY

- 1 Match series type.
- 2 Insert formula.
- 3 Xsolve the equation and check n .

WORKED MODEL

$$\frac{n}{2}(2a + (n - 1)d) = 100.$$

HARD VARIANTS

- 1 Sum equations often become quadratics or exponentials.
- 2 Reject negative or non-integer values of n .
- 3 In context, check whether the answer should be rounded up or exact.

— BOTTOM LINE

Unknown n must be a positive integer.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
sequence or series unknown type	$d = u_{n+1} - u_n, \quad r = \frac{u_{n+1}}{u_n}$
Σ notation	$\sum_{r=1}^n u_r$
given S_n , find u_n	$u_n = S_n - S_{n-1}$
series condition, unknown n	$S_n = \text{given value}$